细菌基因组学

1.上游

1.1 准备

```
# 创建环境
conda create bacteria
# 下载 nextflow
conda install nextflow
# 下载 nf-core
conda install nf-core
nf-core download nf-core/bacass
tar -zxvf nf-core-bacass_2.4.0.tar.gz
#run
nextflow run ./nf-core-bacass_2.4.0/2_4_0 \
 --input samplesheet.csv \
 -profile singularity \
 --outdir outdir \
 --kraken2db 'https://genome-idx.s3.amazonaws.com/kraken/k2_standard_8gb_20210517.tar.gz' \
 --kmerfinderdb /home/wtao/projects/czh/kmerfinder_db/database/bacteria/ \
 --ncbi_assembly_metadata assembly_summary.txt \
 -resume
```

1.2 输入文件

1.2.1 --input samplesheet.csv

ID	R1	R2	LongFastQ	Fast5	GenomeSize
HVKP1	/work/run/projects/wtao/czh/bacteria/SJKP_1.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_1.clean_2.fastq.gz	NA	NA	NA
HVKP17	/work/run/projects/wtao/czh/bacteria/SJKP_17.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_17.clean_2.fastq.gz	NA	NA	NA
HVKP18	/work/run/projects/wtao/czh/bacteria/SJKP_18.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_18.clean_2.fastq.gz	NA	NA	NA
HVKP19	/work/run/projects/wtao/czh/bacteria/SJKP_19.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_19.clean_2.fastq.gz	NA	NA	NA
HVKP20	/work/run/projects/wtao/czh/bacteria/SJKP_20.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_20.clean_2.fastq.gz	NA	NA	NA
HVKP21	/work/run/projects/wtao/czh/bacteria/SJKP_21.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_21.clean_2.fastq.gz	NA	NA	NA
HVKP22	/work/run/projects/wtao/czh/bacteria/SJKP_22.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_22.clean_2.fastq.gz	NA	NA	NA
HVKP23	/work/run/projects/wtao/czh/bacteria/SJKP_23.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_23.clean_2.fastq.gz	NA	NA	NA
CKP28	/work/run/projects/wtao/czh/bacteria/SJKP_28.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_28.clean_2.fastq.gz	NA	NA	NA
HVKP3	/work/run/projects/wtao/czh/bacteria/SJKP_3.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_3.clean_2.fastq.gz	NA	NA	NA
HVKP4	/work/run/projects/wtao/czh/bacteria/SJKP_4.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_4.clean_2.fastq.gz	NA	NA	NA
HVKP5	/work/run/projects/wtao/czh/bacteria/SJKP_5.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_5.clean_2.fastq.gz	NA	NA	NA
HVKP6	/work/run/projects/wtao/czh/bacteria/SJKP_6.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_6.clean_2.fastq.gz	NA	NA	NA
HVKP7	/work/run/projects/wtao/czh/bacteria/SJKP_7.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_7.clean_2.fastq.gz	NA	NA	NA
HVKP8	/work/run/projects/wtao/czh/bacteria/SJKP_8.clean_1.fastq.gz	/work/run/projects/wtao/czh/bacteria/SJKP_8.clean_2.fastq.gz	NA	NA	NA

- `ID` (required): Unique sample IDs, must start with a letter, and can only contain letters, numbers or underscores

- `R1` (optional): Paths to (forward) reads zipped FastQ files
- `R2` (optional): Paths to reverse reads zipped FastQ files, required if the data is paired-end
- `LongFastQ` (optional): Paths to long reads zipped FastQ files
- `Fast5` (optional): Paths to the directory containing FAST5 files
- `GenomeSize` (optional): A number (including decimals) ending with 'm', representing genome size.

1.2.2 --kraken2db /home/wtao/projects/czh/kmerfinder_db/database/bacteria/

- kraken2db：Path to the Kmerfinder bacteria database
- 下载地址：https://bitbucket.org/genomicepidemiology/kmerfinder_db/src/master/

1.2.3 --ncbi_assembly_metadata assembly_summary.txt

- ncbi_assembly_metadata：Master file (*.txt) containing a summary of assemblies available in GeneBank or RefSeq.
- 下载地址：https://ftp.ncbi.nlm.nih.gov/genomes/ASSEMBLY_REPORTS/assembly_summary_refseq.txt

1.2.4 outdir

- 详细说明：https://nf-co.re/bacass/2.4.0/docs/output/
- 重要结果：
  - `Prokka/{ID}/`
    - `*.gff` : Annotation in gff format
    - `*.txt` : Annotation in text format
    - `*.faa` : Protein sequences in fasta format

2. 下游

2.1 基因组比对

```
conda install -c bioconda mummer
nucmer --prefix=genome_comparison ./参考基因组/ncbi_dataset_k2044/ncbi_dataset/data/GCA_000009885.1/GCA_000009885.1_ASM988v1_genomic.fna ./P24052701/bacteria/outdir/Prokka/CKP28/CKP28.fna
#  --prefix=genome_comparison: 设置输出文件的前缀
#  第一个文件路径是参考基因组
```

```
#  第二个文件路径是要比对的目标基因组
#  输出将生成 .delta 文件，包含比对结果
show-coords -q -c -l genome_comparison.delta > genome_comparison.coords
#  将 delta 文件转换为易读的坐标格式:
#  -q: 基于查询序列(query)的排序方式展示比对结果
#  -c: 显示覆盖度信息
#  -l: 显示序列长度
show-diff -q genome_comparison.delta > genome_differences.txt
#  分析并输出基因组间的差异:
#  -q: 基于查询序列的分析模式
#  输出结果包含插入、缺失和重排等结构变异信息
```

2.2 VFDB毒力因子预测

```
#下载blast+
wget ftp://ftp.ncbi.nlm.nih.gov/blast/executables/blast+/LATEST/ncbi-blast-2.16.0+-x64-linux.tar.gz
tar -zxvf ncbi-blast-2.16.0+-x64-linux.tar.gz
mv ncbi-blast-2.16.0+  blast
#export PATH=/work/run/projects/wtao/czh/blast/bin:$PATH
source ~/.bashrc
#下载毒力因子数据库: https://www.mgc.ac.cn/VFs/download.htm
gzip -d VFDB_setB_pro.fas.gz
#构建数据库
makeblastdb -in VFDB_setB_pro.fas -dbtype prot -out vfdb
#与数据库比对
blastp -query P24052701/bacteria/outdir/Prokka/CKP28/CKP28.faa  -db vfdb -out vfdb_results.txt -evalue 1e-5 -outfmt "6 qseqid sseqid qstart qend length qlen slen evalue bitscore stitle" -num_threads 20
```

2.3 COG注释

```
#diamond跟blast+差不多，速度更快
#diamond下载:
wget -c https://github.com/bbuchfink/diamond/releases/download/v2.1.11/diamond-linux64.tar.gz
tar -zxvf diamond-linux64.tar.gz
#COG数据库下载
wget -c https://ftp.ncbi.nlm.nih.gov/pub/COG/COG2024/data/COGorg24.faa.gz
gunzip COGorg24.faa.gz
#run
```

```
./diamond help
../diamond/diamond makedb --in COGorg24.faa -d cog_db
../diamond/diamond blastp -d cog_db -q ../P24052701/bacteria/outdir/Prokka/CKP28/CKP28.faa -out cog_results.txt -e 1e-5 -k 1 -p 20 --outfmt 6 qseqid sseqid qstart qend length qlen slen evalue bitscore stitle
../diamond/diamond blastp \
-d cog_db \
-q ../P24052701/bacteria/outdir/Prokka/CKP28/CKP28.faa \
-o cog_results.txt \
-e 1e-5 \
-k 1 \
-p 20
```

2.4 circos

```
#下载
conda install circos
circos -v
#circos文件地址
conda list circos | grep circos
```

2.4.1 数据预处理:circos_pre.py

```
from Bio import SeqIO
recs = SeqIO.parse("../outdir/outdir/Prokka/HVKP23/HVKP23.gbk", "genbank")
# 使用 SeqIO.parse() 来处理多个记录
forward = []
reverse = []
total = 0
for rec in recs:
    length = len(rec)
    for feature in rec.features:
        if feature.type != 'CDS':
            continue
        if feature.location.strand == 1:
            forward.append((feature.qualifiers['locus_tag'][0], int(feature.location.start) + total, int(feature.location.end) + total))
        elif feature.location.strand == -1:
            reverse.append((feature.qualifiers['locus_tag'][0], int(feature.location.start) + total, int(feature.location.end) + total))
    total += length
```

```
def get_forward_strand(strands):
    """
    Create forward strands
    """
    with open("HVKP23/forward.txt", "w") as f:
        for strand in strands:
            f.writelines("chr1\t%d\t%d\t%s\n" % (strand[1], strand[2], strand[0]))

def get_reverse_strand(strands):
    """
    Create reverse strands
    """
    with open("HVKP23/reverse.txt", "w") as f:
        for strand in strands:
            f.writelines("chr1\t%d\t%d\t%s\n" % (strand[1], strand[2], strand[0]))
def calc_GC_content(seq):
    """
    Return GC content of input sequence
    """
    gc = sum(seq.count(x) for x in ['G','C','g','c'])
    gc_content = gc/float(len(seq))
    return round(gc_content, 4)

def calc_GC_skew(seq):
    """
    Reuturn GC skew of input sequence
    """
    g = seq.count('G') + seq.count('g')
    c = seq.count('C') + seq.count('c')
    try:
        skew = (g - c)/float(g + c)
    except ZeroDivisionError:
        skew = 0
    return round(skew, 4)

window = 10000
step = 5000
seqs = SeqIO.parse("../outdir/outdir/Prokka/HVKP23/HVKP23.fna", "fasta")
seq_total = 0
with open("HVKP23/gc_skew.txt", "w") as f1, open("HVKP23/gc_content.txt", "w") as f2:
    for seq in seqs:
        seq_1 = seq.seq
        seq_len = len(seq_1)
        for i in range(0, len(seq_1), step):
```

```
            subseq = seq_1[i:i+window]
            gc_content = calc_GC_content(subseq)
            gc_skew = calc_GC_skew(subseq)
            i += seq_total
            start = i + 1 if i + 1 <= len(seq_1)+seq_total else i
            end = i + step if i + step <= len(seq_1)+seq_total else len(seq_1)+seq_total
            f1.writelines("chr1\t%d\t%d\t%s\n" % (start, end, gc_skew))
            f2.writelines("chr1\t%d\t%d\t%s\n" % (start, end, gc_content))
        seq_total += seq_len
get_forward_strand(forward)
get_reverse_strand(reverse)
```

2.4.2 circos配置：circos.cnof

plot部分更改配置：文件夹地址（红色标注）等

```
<<include colors_fonts_patterns.conf>>
<ideogram>
  <spacing>
    default  = 0u
    break    = 0u
  </spacing>
  radius    = 0.85r
  thickness  = 5p
  stroke_thickness  = 1
  stroke_color = black
  fill = yes
  fill_color = black
</ideogram>

<image>
dir  = .
file  = circos_HVKP23.png
png  = yes
svg  = yes

# radius of inscribed circle in image
radius        = 1000p

# by default angle=0 is at 3 o'clock position
angle_offset      = -90

#angle_orientation = counterclockwise

```

```
auto_alpha_colors = yes
auto_alpha_steps  = 5
background = white
</image>

karyotype = ./node.txt
chromosome_units = 1000
chromosomes_display_default = yes
show_ticks          = yes
show_tick_labels    = yes
show_grid          = no
grid_start          = dims(ideogram,radius_inner) - 0.5r
grid_end            = dims(ideogram,radius_inner)

# -------------------- ticks start -------------------- #

<ticks>
# if yes, skip of label of 0bp
skip_first_label = no
# if yes, skip of lable of last
skip_last_label = no

# ticks radius to show, outside or inside
radius  = dims(ideogram,radius_outer)

# genome size * label_multiplier = label units
label_multiplier = 1e-6

tick_separation = 2p
min_label_distance_to_edge = 0p
label_separation = 5p
# label distance to ticks
label_offset    = 5p
label_size = 8p
size = 20p

<tick>
spacing        = 10000u
color          = gray
show_label      = no
thickness      = 1p
</tick>

<tick>
spacing        = 100000u
color          = black
show_label      = yes
```

```
label_size      = 16p
thickness      = 2p

# label format, %.1f = 1.1, %d = 1, %.2f = 1.10
format          = %.1f
grid            = yes
grid_color      = white
grid_thickness  = 1p
</tick>

<tick>
spacing        = 1000000u
size            = 30p
color          = black
show_label      = yes
label_size      = 24p
thickness      = 3p
# label format, %.1f = 1.1, %d = 1, %.2f = 1.10
format          = %d
# plus Mb to lables
suffix          = " Mb"
grid            = yes
grid_color      = dgrey
grid_thickness  = 1p
</tick>
</ticks>
# --------------------- ticks ends ---------------------- #

# --------------------- plot start ---------------------- #
<plots>
  <plot>
    type  = tile
    file  = ./HVKP23/forward.txt
    r1    = 0.95r
    r0    = 0.90r
    margin = 1b
    stroke_thickness = 0
    orientation = center
    # tile height
    thickness = 20p
    # different layer index height
    padding = 5p
    layers = 1
    color = orange
  </plot>

  <plot>
```

```
    type  = tile
    file  = ./HVKP23/reverse.txt
    r1    = 0.92r
    r0    = 0.87r
    margin = 1b
    stroke_thickness = 0
    thickness = 20p
    padding = 5p
    # display one layer of plot
    layers = 1
    <rule>
      condition = var(size) > 3kb
      color = black
    </rule>
  </plot>

  <plot>
    type  = tile
    file  = ./HVKP23/feature_diff_k2044.txt
    r1    = 0.85r
    r0    = 0.8r
    margin = 1b
    stroke_thickness = 0
    color = red
    thickness = 20p
    padding = 5p
    # display one layer of plot
    layers = 1
    <rule>
      condition = var(size) > 3kb
      color = black
    </rule>
  </plot>

  <plot>
    type  = tile
    file  = ./HVKP23/feature_diff.txt
    r1    = 0.78r
    r0    = 0.73r
    margin = 1b
    stroke_thickness = 0
    color = blue
    thickness = 20p
    padding = 5p
    # display one layer of plot
    layers = 1
    <rule>
```

```
        condition = var(size) > 3kb
        color = black
      </rule>
    </plot>

    <plot>
      type = line
      r1 = 0.70r
      r0 = 0.60r
      file = ./HVKP23/gc_skew.txt
      thickness = 1
      max_gap = 1u
      color  = vdgrey
      # max/min value of line plotting
      min    = -0.15
      max    = 0.15
      fill_color = vdgrey_a3
      <backgrounds>
        <background>
          color    = vvlgreen
          y0        = 0.05
        </background>
        <background>
          color    = vvlred
          y1        = -0.05
        </background>
      </backgrounds>

      <axes>
        <axis>
          color    = lgrey_a2
          thickness = 1
          spacing  = 0.04r
        </axis>
      </axes>
    </plot>

    <plot>
      type = histogram
      r1 = 0.55r
      r0 = 0.45r
      file = ./HVKP23/gc_content.txt
      thickness = 1
      max_gap = 1u
      min    = 0
      max    = 1
      <rules>
```

```
      <rule>
        condition = 1
        fill_color = eval(sprintf("spectral-9-div-%d", remap_int(var(value), 0,1,1,9)))
      </rule>
    </rules>
  </plot>
</plots>

<<include housekeeping.conf>>
```

2.4.3 run

```
circos -conf circos.cnof
```

2.5 R整理

2.5.1 process_genome_data函数

```
process_genome_data <- function(base_path) {
  library(data.table)
  library(dplyr)

  # Read genome differences
  genome_diff <- fread(file.path(base_path, "genome_differences.txt"), fill = TRUE, skip = 4)
  colnames(genome_diff) <- c("locus", "type", "start", "end", "len", "len_ref", "diff")

  # Find GFF file
  gff_file <- list.files(base_path, pattern = "\\.gff$", full.names = TRUE)
  if (length(gff_file) == 0) stop("No GFF file found.")
  gff_lines <- readLines(gff_file[1])
  header_indices <- grep("^##", gff_lines)
  gene_indices <- tail(header_indices, 2)
  gff_data <- gff_lines[(gene_indices[1] + 1):(gene_indices[2] - 1)]

  extract_gene_info <- function(data) {
    do.call(rbind, lapply(data, function(line) {
      if (grepl("\\bgene\\b", line)) {
        fields <- strsplit(line, "\t")[[1]]
        locus <- fields[1]
```

```
        start <- as.numeric(fields[4])
        end <- as.numeric(fields[5])
        gene_match <- regmatches(line, regexec("gene=([^;]+)", line))
        gene <- ifelse(length(gene_match[[1]]) > 1, gene_match[[1]][2], NA)
        protein_match <- regmatches(line, regexec("locus_tag=([^;]+)", line))
        protein <- ifelse(length(protein_match[[1]]) > 1, protein_match[[1]][2], NA)
        return(data.frame(locus = locus, start = start, end = end, gene = gene, protein = protein))
      }
      return(NULL)
    }))
  }

  gene_info <- extract_gene_info(gff_data)

  # Find GBK file
  gbk_file <- list.files(base_path, pattern = "\\.gbk$", full.names = TRUE)
  if (length(gbk_file) == 0) stop("No GBK file found.")
  gbk_lines <- readLines(gbk_file[1])

  locus_lines <- gbk_lines[grep("LOCUS", gbk_lines)]
  locus_info <- list()
  for (i in 1:length(locus_lines)) {
    matches <- regmatches(locus_lines[i], regexec("LOCUS\\s+(\\S+)\\s+(\\d+)", locus_lines[i]))
    locus <- matches[[1]][2]
    length <- matches[[1]][3]
    locus_info[[i]] <- data.frame(locus = locus, length = length)
  }

  cumulative_lengths <- do.call(rbind, locus_info)
  cumulative_lengths$length <- c(0, head(cumulative_lengths$length, -1)) %>% as.numeric()
  cumulative_lengths$length <- cumsum(cumulative_lengths$length)
  merged_gene_info <- merge(gene_info, cumulative_lengths, by = "locus")
  merged_gene_info$start <- merged_gene_info$start + merged_gene_info$length
  merged_gene_info$end <- merged_gene_info$end + merged_gene_info$length

  genome_diff$locus <- as.character(genome_diff$locus)
  merged_diff_info <- merge(genome_diff, cumulative_lengths, by = "locus")
  merged_diff_info$start <- merged_diff_info$start + merged_diff_info$length
  merged_diff_info$end <- merged_diff_info$end + merged_diff_info$length
  merged_diff_info <- within(merged_diff_info, {
    swap_needed <- end < start
    temp <- ifelse(swap_needed, start, end)
    start <- ifelse(swap_needed, end, start)
```

```
    end <- temp
  })

  # Remove temporary columns
  merged_diff_info$swap_needed <- NULL
  merged_diff_info$temp <- NULL
  merged_diff_info <- as.data.frame(merged_diff_info)
  feature_diff <- merged_diff_info[c("start", "end", "type")]
  colnames(feature_diff) <- c("#start", "end", "label")
  feature_diff <- cbind(chr = "chr1", feature_diff)
  write.table(feature_diff, file = file.path(base_path, "feature_diff.txt"), quote = FALSE, row.names = FALSE, col.names = FALSE, sep = "\t")

  # Merge data
  result <- merged_diff_info %>%
    rowwise() %>%
    mutate(
      type = case_when(
        type == "GAP" & diff > 0 ~ "GAP-insertion",
        type == "GAP" & diff < 0 ~ "GAP-deletion",
        TRUE ~ type
      ),
      gene = paste(
        merged_gene_info$gene[merged_gene_info$start <= end & merged_gene_info$end >= start],
        collapse = ","
      )
    )

  # Construct list
  gene_list <- result %>%
    ungroup() %>%
    group_by(type) %>%
    summarise(genes = paste(unique(gene[gene != ""]), collapse = ",")) %>%
    { setNames(as.list(.$genes), .$type) }
  gene_list <- lapply(gene_list, function(x) strsplit(x, ",")[[1]])

  feature_diff_gene <- merged_gene_info[merged_gene_info$gene %in% unlist(gene_list), ]

  # Read and filter VFDB results
  #vfdb_file <- file.path(base_path, "vfdb_results.txt")
  #vfdb_data <- fread(vfdb_file)
  #siderophore_subset <- subset(vfdb_data, grepl("siderophore", vfdb_data$V10))
  #feature_diff_gene <- feature_diff_gene[feature_diff_gene$protein %in% siderophore_subset$V1, ]
```

```
  feature_diff_gene2 <- feature_diff_gene[c("start", "end", "gene", "protein")]
  colnames(feature_diff_gene2)[1:3] <- c("#start", "end", "label")
  feature_diff_gene2 <- cbind(chr = "chr1", feature_diff_gene2)
  write.table(feature_diff_gene2, file = file.path(base_path, "feature_diff_gene.txt"), quote = FALSE, row.names = FALSE, col.names = FALSE, sep = "\t")
}

```

2.5.2 run

```
process_genome_data("./outdir/Prokka/CKP28/")
```